Investments Portfolio

BUS 315 Investments, Simon Fraser University
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This is a selection of the four assignments I completed in BUS 315. Each one starts from real market data, which I cleaned and analyzed in Excel, and ends with a conclusion I had to defend. The numbers below come straight from my workbooks.

The four assignments at a glance

Assignment	What I did	Main result
1. Gold, stocks and inflation	Compared 56 years of gold returns, stock returns and CPI inflation, 1970 to 2025.	Gold beat inflation in 30 of 56 years but swung far more than stocks.
2. Market history and investing rules	Tax-deferred versus taxable growth, missing the best or worst months, lump sum versus monthly investing (5,000 simulated years).	Missing the 12 best months of 98 years cut the final value of $1 from $13,418 to $1,338.
3. Portfolio optimization and hedging	Built optimal portfolios from 7 securities and 2,766 trading days. Designed a hedge for a concentrated Amazon position.	The lowest-risk portfolio had 17% less daily volatility than SPY (0.93% versus 1.12%).
4. Size and value premiums, rates, bank risk	CAPM regressions on 15 portfolios and 684 months of data. Compared forward rates with actual rates. Measured a duration gap at BMO.	Value stocks (BM5) earned a CAPM alpha of 0.26% a month; the market's rate forecast missed by about 1 to 2 points.

Tools and methods: Excel (regression output, constrained optimization, covariance matrices, Monte Carlo simulation, pivot tables, charts), CAPM and single-index models, efficient frontier, Sharpe ratio, forward rates, duration and dollar duration.

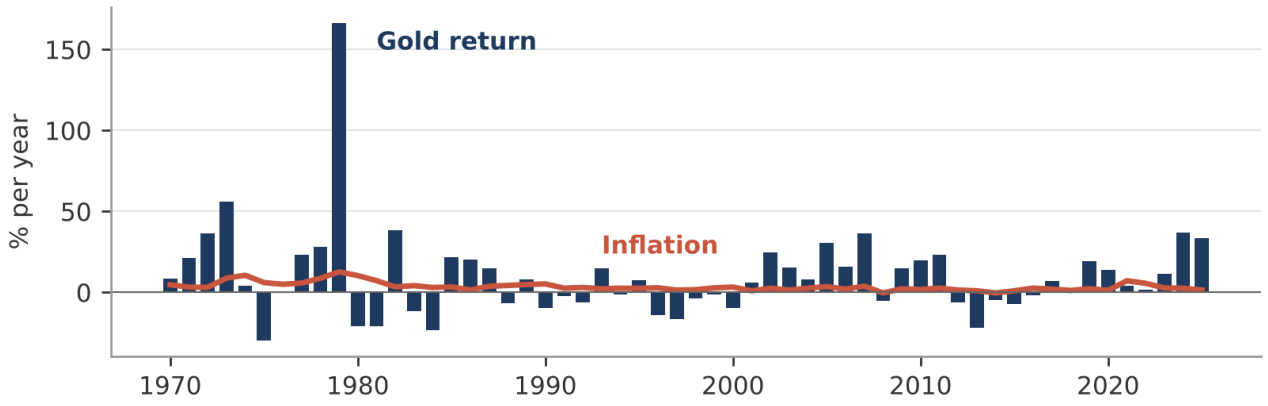
Teamwork: Assignments 2 and 3 were done as a pair with Gautam Upadhyaya.

Assignment 1: Gold, stocks and inflation

Data: monthly stock market returns, gold prices and CPI, January 1970 to August 2025 (668 months).

Many investors hold gold as protection against inflation. I tested that idea with the data.

- **Return and risk.** Gold averaged 0.86% a month with a standard deviation of 5.67%. The stock market averaged 0.95% with a standard deviation of 4.55%. Gold gave a lower return for more risk.
- **Against inflation.** Average annual gold return was 10.1%, against 3.6% average inflation. Gold still beat inflation in only 30 of the 56 years. A single year, 1979 at +166%, pulls the average up a lot.
- **Link to inflation.** Month by month, the correlation between gold returns and inflation was only 0.12. Gold is a loose hedge at best.
- **Real price.** After adjusting for CPI, the gold price rose about 13 times, from 0.92 in January 1970 to 11.83 in August 2025.



Annual gold return (bars) and inflation (line), 1970 to 2025. Source: my Assignment 1 workbook.

Assignment 2: Market history and investing rules

Data: monthly U.S. market and risk-free returns, January 1927 to December 2024 (1,176 months). Done with Gautam Upadhyaya.

Does the tax treatment matter?

I grew $100,000 for 10 years at 8% a year in two accounts. In the tax-deferred account, the 20% tax is paid once at the end. In the taxable account, the tax is paid every year, so the money grows at 6.4%.

Account	Value after 10 years
Tax paid once, at the end	$192,714
Tax paid every year	$185,959
Difference	$6,755

What happens if you miss the best or worst months?

One dollar invested in the market in 1927 grew to $13,418 by 2024. I removed the 12 best months and then the 12 worst months and compared.



Value of $1, 1927 to 2024, on a log scale. Twelve months out of 1,176 decide most of the result.

Lump sum or monthly investing?

I simulated 5,000 one-year outcomes using the market's historical mean (0.95% a month) and standard deviation (5.33%). In each trial I compared investing $1,200 on day one with investing $100 a month.

	Lump sum	Monthly ($100)
Average ending value	$1,345	$1,278
Worst trial	$677	$884

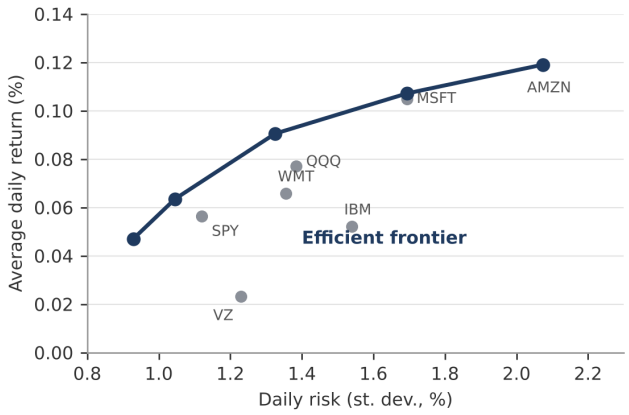
Lump sum finished ahead in about 67% of trials which fits a market that rises more often than it falls. Monthly investing had the better worst case, so it reduces the chance of a bad outcome in exchange for a lower average.

Assignment 3: Portfolio optimization and hedging

Data: daily returns for MSFT, IBM, WMT, VZ, SPY, AMZN and QQQ, January 2015 to December 2025 (2,766 trading days). Done with Gautam Upadhyaya.

I estimated each security's average return, standard deviation and the full covariance matrix, then solved for portfolios under different goals.

Goal	Weights	Daily return	Daily risk	Sharpe
Lowest risk	IBM 3.6%, WMT 22.5%, VZ 34.3%, SPY 39.5%	0.047%	0.93%	0.051
Highest Sharpe ratio	MSFT 33.2%, IBM 5.3%, WMT 38.2%, AMZN 23.3%	0.091%	1.26%	0.072
Single-index model	MSFT 38.1%, WMT 35.6%, AMZN 26.4%	0.095%	n/a	0.074



Efficient frontier from five optimized portfolios. Single securities sit below it.

What I took from it

- The lowest-risk portfolio had about 17% less daily volatility than SPY alone (0.93% versus 1.12%).
- The optimizer left AMZN out of the lowest-risk mix. It carried the highest risk of the seven (2.07%).
- The single-index model gave the same kind of answer as the full covariance model, with fewer inputs. Betas ran from 0.42 (VZ) to 1.19 (MSFT).

Hedging a concentrated position

The scenario: $10 million of capital, with $7 million fixed in Amazon (30,327 shares at $230.82). I looked for the long and short positions in the other six securities that cut the variance of the whole portfolio the most.

- The best hedge shorted QQQ (about 96% of capital) and held long positions in SPY (about 25%), IBM and VZ, plus a small MSFT short. QQQ and AMZN have a correlation of 0.75, which is why it works.
- Portfolio variance fell from 4.30×10^{-4} (Amazon alone) to 0.86×10^{-4}. Daily risk dropped from 2.07% to 0.93%, a cut of about 55%.

- **Limits.** The weights rely on historical correlations, and those can change quickly. The model also ignores the cost of borrowing shares to short. I would reduce the QQQ short a little, accept slightly more variance, and end up less dependent on large-cap tech moving together.

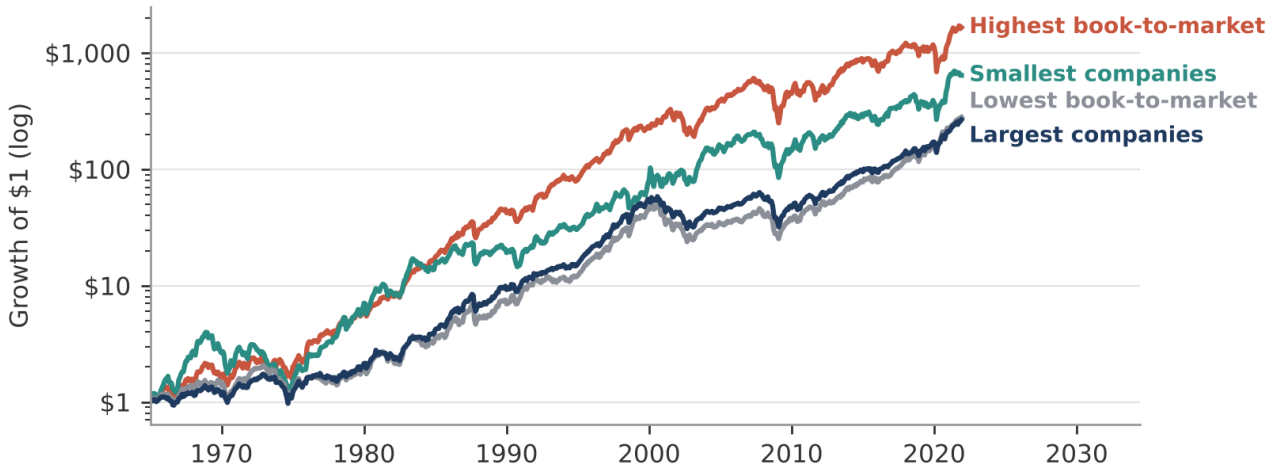
Assignment 4: Size and value premiums, interest rates, bank risk

Part 1. Do small and value stocks earn more?

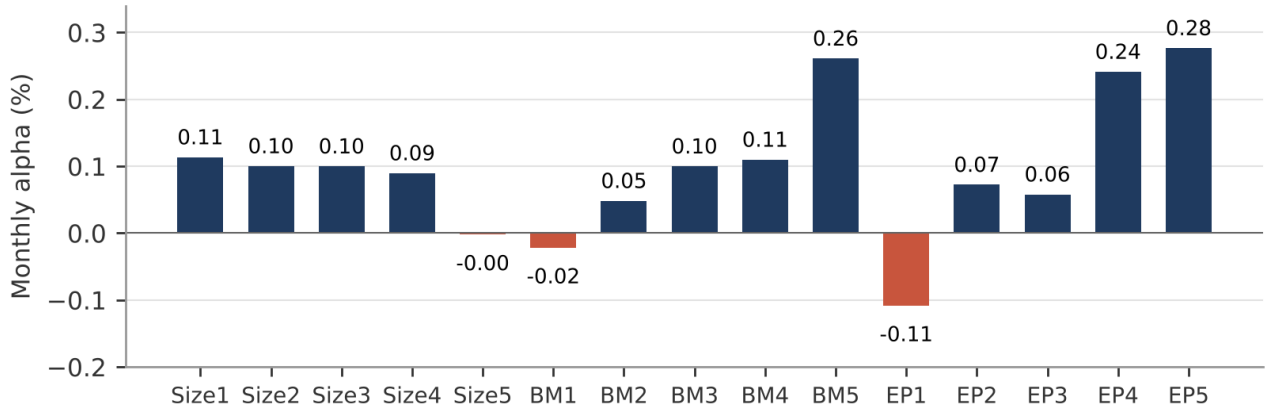
Data: monthly returns for 15 portfolios (5 by size, 5 by book-to-market, 5 by earnings-to-price), January 1965 to December 2021 (684 months).

	Avg. monthly return	St. deviation	CAPM alpha
Smallest companies (Size1)	1.16%	6.4%	+0.11%
Largest companies (Size5)	0.91%	4.3%	0.00%
Highest book-to-market (BM5)	1.24%	5.5%	+0.26%
Lowest book-to-market (BM1)	0.94%	4.8%	−0.02%

- **Size and value premiums.** Small companies and value stocks earned more, and they were also riskier. Small companies were about 50% more volatile than large ones.
- **CAPM did not explain it all.** I ran a CAPM regression for each of the 15 portfolios. If CAPM held perfectly, every alpha would be zero. Small and value portfolios had positive alphas, while the largest and the growth portfolios were at or below zero.
- **The value premium comes and goes.** A long-short portfolio (BM5 minus BM1) earned 0.30% a month overall. It lost 0.17% a month during the 1998 to 2000 dot-com boom, then made 1.11% a month from 2001 to 2004 after growth stocks fell.



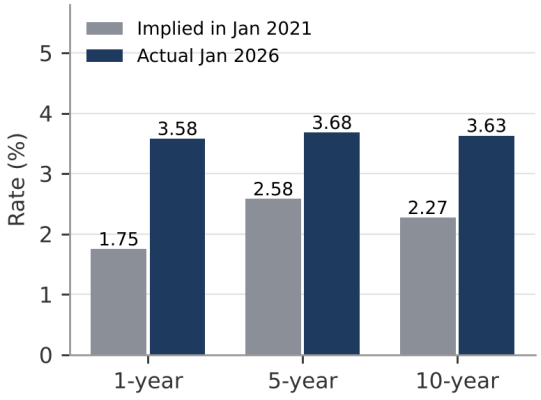
Growth of $1 invested in January 1965 (log scale). Highest book-to-market reached $1,654, smallest companies $640, lowest book-to-market $280, largest companies $266.



CAPM alpha by portfolio, as a percentage per month.

Part 2. Interest rates and bank risk

Could the market predict rates?
Using the 2021 yield curve, I calculated the rates it implied for January 2026 and compared them with the actual rates. The market was wrong: actual rates were roughly 1 to 2 points higher at every maturity.



How much interest-rate risk does BMO carry?
I used BMO's assets and liabilities by maturity band, took the midpoint of each band as its duration, and gave items with no maturity a duration of zero.

- Dollar duration of assets: 3.44 million million-years.
- Dollar duration of liabilities: 1.15 million million-years.
- Gap: 2.29 million million-years, so a 100 basis point rise in rates would cost shareholders about $22.9 billion.
- BMO's reported impact was $1.483 billion. My estimate is much larger because the model is simple: it uses band midpoints and ignores hedges.